

Haluk Yuzukirmizi

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EDUCATION

University of Massachusetts Amherst

Expected Graduation : May 2027

B.S. in Computer Science | Business Minor | Commonwealth Honors College

GPA: 3.95

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Software Engineering, Algorithms, Computer Systems, Multivariate Calculus, Discrete Math, Statistics, Linear Algebra
- **Awards:** Adobe Computer Science Scholarship, Massachusetts High Demand Scholarship, Dean's List (All Semesters)

Break Through Tech AI (Cornell Tech)

May 2026 - May 2027

Artificial Intelligence / Machine Learning Fellowship

EXPERIENCE

IBM

May 2026 - Present

Software Engineering Intern

Austin, TX

- Developing multi-agent AI assistant with IBM watsonx Orchestrate to support the application owners of 95 enterprise applications across IBM's CIO organization, serving **25,000+ internal users**.

Commonwealth of Massachusetts - Executive Office of Technology Services & Security

Apr 2025 - Aug 2025

Artificial Intelligence Intern

Amherst, MA

- Built a serverless multi-modal RAG AI assistant on AWS (Bedrock, Lambda, S3) delivering 24/7 automated Unity High-Performance Computing Platform support to **2,000+ researchers** via Slack.
- Automated knowledge base ingestion across 160+ docs and expanded coverage **30%** by generating Q&A pairs from resolved Slack threads via LLM.
- Reduced average response time **46%** by engineering contextual chunking with document-level metadata injection, enabling better semantic search and pre-search filtering.
- Cut computing costs **12.3×** via metadata filtering and prompt engineering while maintaining **95% model accuracy**.

RESEARCH

University of Massachusetts Amherst | Undergraduate Research Assistant

Amherst, MA

Advanced Classical and Quantum Information Research Lab | PI: Prof. Don Towsley

Feb 2026 - Present

- Developing a Python architecture to extend Quantum Walk Neural Networks (QWNNs) with multi-walker entanglement, investigating its effect on learning discrete-time quantum walks across complex graph topologies.

Department of Mechanical and Industrial Engineering | Mentor: Prof. Saketh Sridhara

Feb 2026 - Present

- Benchmarking KAN vs. CNN architectures via neural reparameterization for structural topology optimization, targeting material volume minimization under stress constraints.

Computer Vision Lab | Mentor: Prof. Subhransu Maji

Sept 2024 - May 2025

- Conducted research on Detector-based Importance Sampling for Counting (DISCount), a human-in-the-loop machine learning model for object counting in high-resolution images.
- Built a modular importance sampling pipeline for high-resolution object counting with imperfect detectors, achieving **90%+ accuracy** with only **20% of tiles labeled**, outperforming triangle and uniform distribution baselines and cutting labeling costs by **4×**.

PROJECTS

[hyuzukirmizi.github.io](https://github.com/hyuzukirmizi)

DoorSmash (HackUMass XIII, 36 hours) | TypeScript(React), Python(FastAPI), Supabase(PostgreSQL)

[DoorSmash](#)

- Launched a full-stack nutrition platform for campus meal ordering and health tracking, used by **500+ students**.
- Built an agentic AI chatbot (Pydantic AI, Gemini, ElevenLabs) handling conversational ordering, nutrition tracking, and personalized health recommendations via **20+ tool calls**.

SKILLS

Programming Languages: Python, JavaScript/TypeScript, C, Java, Rust, C++, SQL, Go

Technologies: AWS, Docker, REST API, LangChain, TensorFlow, Pandas, Numpy, Matplotlib, PyTorch